

a

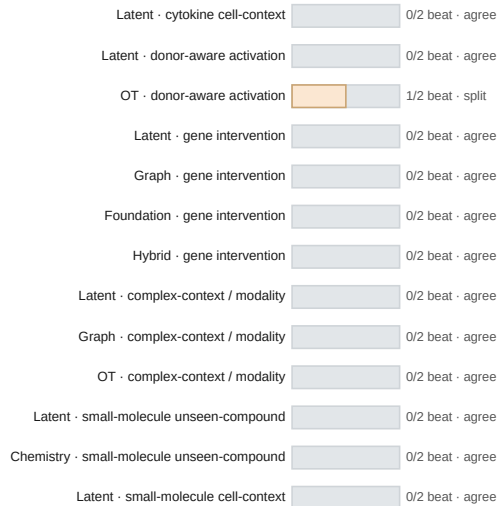
Task × family fit verdict

method family

	Latent	Graph	Foundation	Hybrid	Optimal transport	Chemistry
Cytokine cell-context (Kang)	simple baseline best +0.08	n/e	n/e	n/e	n/e	n/e
Donor-aware activation (Soskic)	simple baseline best +0.05	n/e	n/e	simple baseline best +0.04	CellIoT +0.22 single-method	n/e
Gene intervention (primary-T CRISPR)	simple baseline best -0.32	simple baseline best -0.18	simple baseline best -0.34	simple baseline best -0.40	simple baseline best +0.06	n/e
Complex-context / modality (Frangieh)	simple baseline best +0.07	simple baseline best +0.04	n/e	simple baseline best -0.45	simple baseline best +0.12	n/e
Small-molecule unseen-compound (OP3)	simple baseline best -0.00	n/e	n/e	simple baseline best -0.16	simple baseline best +0.01	simple baseline best +0.00
Small-molecule cell-context (OP3)	simple baseline best +0.03	n/e	n/e	simple baseline best -0.09	simple baseline best +0.11	FP-ridge +0.24 single-method

verdict (response-direction Pearson- Δ vs universal simple baseline)
simple baseline (tie → simplicity)
single-method exception
not evaluated

b Within-family agreement



0 1 2

members evaluated (per family × task)

members evaluated
members beating baseline